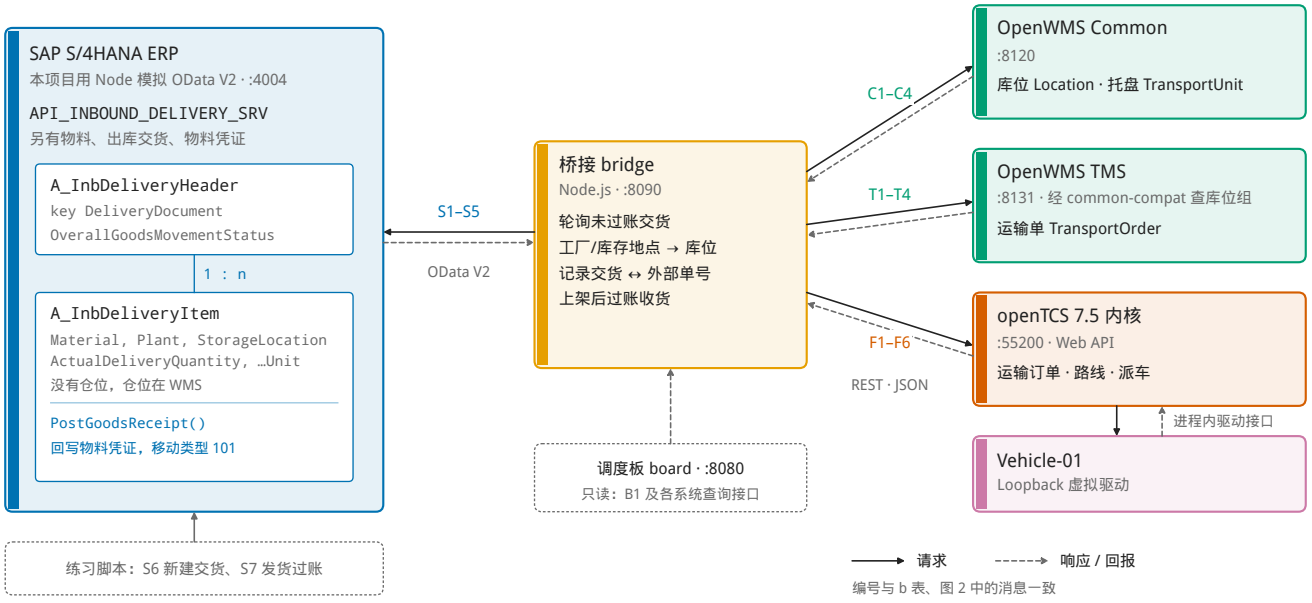
a




b

| 编号 | 方法 | 路径 | 作用 | 阶段 |
|---|---|---|---|---|
| SAP ERP · {INB} = /sap/opu/odata/sap/API_INBOUND_DELIVERY_SRV;v=0002 | | | | |
| {PROD} = API_PRODUCT_SRV, {OUT} = API_OUTBOUND_DELIVERY_SRV;v=0002; 过账需 X-CSRF-Token 与 If-Match | | | | |
| S1 | GET | {INB}/$metadata | 就绪探测，读元数据 | ① |
| S2 | GET | {INB}/A_InbDeliveryHeader?$filter=…Status eq 'A' | 轮询未过账入库交货 | ② |
| S3 | GET | {INB}/A_InbDeliveryHeader('1800000001') | 读交货状态与 ETag | ⑥ |
| S4 | HEAD | {INB}/ X-CSRF-Token: Fetch | 取 CSRF 令牌 | ⑥ |
| S5 | POST | {INB}/PostGoodsReceipt?DeliveryDocument='1800000001' | 过账收货，移动类型 101 | ⑥ |
| S6 | GET | {PROD}/A_Product A_ProductStorageLocation | 同步物料和库存地点 | ① |
| S7 | POST | {OUT}/PostGoodsIssue?DeliveryDocument='8000000001' | 发货过账，本链路不自动调用 | 出库 |
| OpenWMS Common · http://openwms-common:8120 | | | | |
| C1 | GET | /actuator/info | 就绪探测 | ① |
| C2 | GET POST | /v1/locations?locationId=… | 查询 / 创建库位 | ① |
| C3 | GET POST | /v1/transport-units?bk=…[&actualLocation=…&tut=PALLET] | 查询 / 创建托盘 | ① |
| C4 | PATCH | /v1/transport-units?bk=…&newLocation=… | 移动托盘 | ③ ⑥ |
| OpenWMS TMS · http://openwms-tms:8131 | | | | |
| T1 | GET | /actuator/info | 就绪探测 | ① |
| T2 | POST | /v1/transport-orders | 创建运输单，返回 201 + Location | ③ |
| T3 | GET | /v1/transport-orders/{pKey} | 读运输单状态 | ③ ⑥ |
| T4 | POST | /v1/transport-orders/{pKey}?state=STARTED\|FINISHED\|ONFAILURE\|CANCELED | 改运输单状态 | ③ ⑥ |
| openTCS · http://opentcs:55200 | | | | |
| F1 | GET | /v1/kernel/version | 就绪探测 | ① |
| F2 | GET PUT | /v1/plantModel | 读取 / 上传工厂模型 | ① |
| F3 | PUT | /v1/vehicles/Vehicle-01/commAdapter/… | 挂 Loopback 驱动并允许接单 | ① |
| F4 | POST | /v1/transportOrders/INB-{DeliveryDocument} | 下上架订单：收货位装货 → 存储位卸货 | ④ |
| F5 | GET | /v1/transportOrders/INB-{DeliveryDocument} | 读订单状态 | ⑤ |
| F6 | POST | /v1/transportOrders/INB-{DeliveryDocument}/withdrawal?immediate=false | 失败时撤回订单 | 失败 |
| 桥接 · http://bridge:8090 | | | | |
| B1 | GET | /status /health | 进度、对照 links、日志 | 观测 |

阶段：① 初始化 ② 查询交货 ③ 仓库下单 ④ 车队下单 ⑤ 执行监控 ⑥ 收货过账（与图 2 对应）。车辆与 openTCS 之间是进程内驱动接口，没有 HTTP 端点。